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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID:

1. Topic (12 words max)

"PaddyGuard AI – Smart Segmentation & Hybrid Prediction for Rice Diseases"

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Paddy farming is a cornerstone of Sri Lanka's agricultural economy, yet crop diseases such as rice blast, bacterial blight, and brown spot continue to cause significant yield losses, estimated to reach up to 30% annually in severe outbreaks [1]. Early and accurate disease identification is critical for effective intervention; however, current practices rely heavily on farmers' visual inspection and experience, which are often unreliable during early infection stages. This results in delayed or incorrect treatment, increased pesticide misuse, environmental degradation, and financial loss for smallholder farmers.

Recent advances in artificial intelligence (AI), particularly deep learning-based image classification, have demonstrated promising results in plant disease detection [2], [3]. Nevertheless, most existing solutions focus on **whole-image classification**, ignoring fine-grained disease patterns such as lesion shape, texture variation, and spatial progression. This limits diagnostic accuracy under real-field conditions where lighting variations, background noise, and overlapping symptoms are common. Furthermore, many AI models function as "black boxes," offering predictions without explanations, which reduces farmer trust and adoption.

Another critical limitation is the lack of **integrated decision support**. Current systems typically address disease detection, treatment recommendation, farmer education, and yield forecasting as isolated components. This fragmentation forces farmers to rely on multiple tools, many of which require high computational resources, continuous internet connectivity, or are unsuitable for mobile deployment in rural contexts.

This research addresses these challenges by proposing **PaddyGuard AI**, an integrated, segmentation-driven hybrid AI system for paddy disease analysis. By prioritizing optimized semantic segmentation to isolate disease-affected regions, followed by interpretable feature analysis, hybrid DL–ML disease prediction, and yield impact forecasting, the proposed solution enables early diagnosis, explainable predictions, and practical decision support. The system is designed to be lightweight, mobile-compatible, and suitable for real-world deployment, directly supporting Sustainable Development Goal 2 (Zero Hunger).

References

[1] International Rice Research Institute, “Rice Blast Disease,” IRRI, 2023.

Available: https://rice-diseases.irri.org/
[2] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.

Available: https://www.sciencedirect.com/science/article/abs/pii/S0168169917311742?
[3] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, Art. no. 1419, 2016.

Available: https://www.frontiersin.org/articles/10.3389/fpls.2016.01419/full

6. Brief description of Existing Research and Systems (300 words max)

Existing research in AI-based plant disease detection primarily focuses on convolutional neural network (CNN) architectures trained on large-scale leaf image datasets such as PlantVillage [4]. These systems report high accuracy under controlled conditions but often fail to generalize to real-field environments due to background noise, illumination variation, and occluded disease symptoms. Transfer learning approaches using pre-trained models such as VGG, ResNet, and EfficientNet have been explored to reduce training cost and improve performance [5].

More recent studies have introduced semantic segmentation models (e.g., U-Net, Mask R-CNN) to localize disease-affected regions before classification [6]. While segmentation improves accuracy, many approaches lack optimization strategies to handle class imbalance and over-segmentation, especially for paddy crops with visually similar disease patterns. Additionally, segmentation outputs are rarely leveraged for deeper disease analysis; instead, they are used merely as a preprocessing step.

Hybrid deep learning–machine learning
(DL–ML) models have been proposed to improve interpretability by combining deep feature extraction with classical classifiers such as SVM and Random Forest [7]. However, these hybrids typically operate on whole-image features rather than segmented lesion-level information, limiting their ability to explain disease etiology. Furthermore, explainable AI (XAI) techniques such as SHAP and LIME remain underutilized in agricultural disease systems.

Yield impact analysis and decision support are addressed separately in agricultural forecasting systems using statistical or ML-based regression models [8]. These systems are rarely integrated with disease detection pipelines, preventing holistic "what-if" analysis for farmers.

In summary, existing systems suffer from fragmentation, limited interpretability, insufficient segmentation optimization, and lack of integration across diagnosis, explanation, and yield forecasting. The proposed research differentiates itself by introducing a **segmentation-first, hybrid, and explainable AI pipeline**, specifically tailored for paddy disease management in real-field and mobile deployment contexts.

References

[4] S. Sladojevic *et al.*, "Deep neural networks based recognition of plant diseases," *Computational Intelligence and Neuroscience*, 2016.

Available: https://ojs.kmutnb.ac.th/index.php/ijst/article/view/
[5] J. Amara *et al.*, "A deep learning-based approach for banana leaf diseases classification," *ICIAR*, 2017.

Available: https://www.btw2017.informatik.uni-stuttgart.de/slidesandpapers/E1-10/paper_web.pdf
[6] A. Ronneberger *et al.*, "U-Net: Convolutional networks for biomedical image segmentation," *MICCAI*, 2015.

Available: https://arxiv.org/abs/1505.04597?
[7] X. Zhang *et al.*, "Hybrid CNN–SVM models for plant disease classification," *IEEE Access*, 2021.
[8] M. Kamilaris and F. Prenafeta-Boldú, "Deep learning in agriculture," *Computers and Electronics in Agriculture*, 2018.

Available: https://colab.ws/articles/10.1016/j.compag.2018.02.016?u

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, **PaddyGuard AI**, is an integrated AI-based decision support system designed to provide early paddy disease detection, interpretable diagnosis, and yield impact forecasting for smallholder farmers. The system directly addresses the gaps identified in existing research by adopting **segmentation-first hybrid AI architecture** and unifying multiple agricultural support functions into a single mobile-compatible platform.

How the Solution Addresses Identified Gaps

- Introduces optimized semantic segmentation to isolate disease-affected regions instead of relying on whole-image classification.
- Utilizes hybrid DL–ML models to balance prediction accuracy and interpretability.
- Applies explainable AI (XAI) techniques to justify predictions using lesion-level features.
- Integrates disease prediction with yield impact forecasting for practical decision-making.
- Designed for lightweight deployment suitable for low-resource mobile devices.

System Perspective (IPO Model)

Input

- Paddy leaf images captured via mobile devices
- Climate and environmental data (temperature, rainfall, humidity)
- Historical disease and yield datasets

Process

1. **Optimized Semantic Segmentation**
   Leaf images are processed using an optimized U-Net-based segmentation model refined via Bayesian optimization to accurately isolate disease lesions.
2. **Feature Extraction and Analysis**

   Segmented regions are analyzed to extract color, texture, and shape features using both handcrafted methods (GLCM, HSV histograms) and deep encoders.
3. **Hybrid Disease Prediction**

   Deep features are fused with handcrafted features and classified using hybrid DL–ML models (EfficientNet + XGBoost).
4. **Explainable AI Layer**

   SHAP/LIME techniques explain model decisions by highlighting influential lesion features.

5. **Yield Impact Forecasting**

   Sequential disease severity and climate data are processed using an LSTM–XGBoost hybrid model to forecast yield loss.

6. **Decision Support Output**

   Results are presented through a mobile interface with disease diagnosis, explanations, and actionable insights.

Output

- Disease classification with confidence score
- Visual explanation of disease symptoms
- Predicted yield impact
- Farmer-oriented decision support information

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The project team will acquire the necessary **agricultural domain knowledge**, **technical expertise**, and **data resources** through structured academic and practical approaches. Domain expertise in paddy diseases will be developed by reviewing agricultural research publications, consulting extension service materials from organizations such as IRRI, and analyzing symptom descriptions from authoritative datasets. This theoretical knowledge will be supplemented by interpreting real disease images used in model training.

Technical expertise will be gained through coursework in machine learning, software engineering, and AI, complemented by hands-on experimentation using TensorFlow, PyTorch, scikit-learn, and OpenCV. Each team member is responsible for conducting **component-specific self-research**, including literature review and prototype development, ensuring deep individual technical competence.

The primary data sources will be publicly available **Kaggle datasets**, including rice leaf disease image datasets and tabular yield–climate datasets. Where segmentation masks are unavailable, bounding boxes will be converted into masks using annotation tools. Data preprocessing and augmentation will be applied to enhance robustness under real-field conditions.

This project does **not require ethical clearance**, as it exclusively uses publicly available datasets and does not involve human subjects, personal data, or field experiments involving farmers.

9. Objectives and Novelty

Main Objective

To design and evaluate an integrated, segmentation-driven hybrid AI system for accurate, interpretable, and practical paddy disease diagnosis and yield impact forecasting.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22273680	Improve lesion-level segmentation accuracy for paddy diseases.	• Implement U-Net / SegFormer models • Apply Bayesian optimization for hyperparameters • Evaluate using IoU, Dice score	Disease-specific segmentation optimization for real-field paddy images.
IT22168740	Extract interpretable lesion features for disease differentiation.	• Handcrafted + deep feature extraction • Feature clustering and rule-based inference • Feature importance evaluation	Lesion-level etiology inference using hybrid feature fusion.
IT22303820	Achieve high-accuracy and explainable disease classification.	• EfficientNet feature extraction • XGBoost/SVM classification • SHAP-based explanation analysis	Explainable hybrid classification using segmented features.

IT22273680	Forecast yield loss using disease progression and climate data.	• Time-series modeling with LSTM • Regression using XGBoost • Scenario-based forecasting	Integrated disease-to-yield forecasting pipeline.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Find the datasets

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Bhagya Nathali	Silva	[signature]
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				